Daily Combinatorics Drill #4

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Binomial Theorem · Pascal's Rule · Coefficient Extraction · Substituting Values into Expansions.

Section A Rapid Recognition (≤15 s each)

1. Coefficient of x^2 in $(1+x)^5$.

Ans: $\binom{5}{2} = 10$

2. Coefficient of x in $(x+2)^3$.

Ans: 12

3. Row 5 of Pascal's triangle.

Ans: 1, 5, 10, 10, 5, 1

4. Coefficient of x^3y^2 in $(x+y)^5$.

Ans: $\binom{5}{2} = 10$

5. Sum of all coefficients of $(1+x)^7$.

Ans: $2^7 = 128$

6. Constant term of $\left(x + \frac{1}{x}\right)^4$.

Ans: $\binom{4}{2} = 6$

7. Express $\binom{9}{4} + \binom{9}{5}$ as one coefficient.

Ans: $\binom{10}{5}$ (which is 252)

8. $\binom{6}{0} - \binom{6}{1} + \binom{6}{2} - \cdots + \binom{6}{6}$.

Ans: 0

9. True or false: coefficient of x^2 in $(1-x)^{10}$ is negative.

Ans: False — it is +45

10. Largest coefficient in $(1+x)^4$.

Ans: $\binom{4}{2} = 6$

Section B Manipulation Drills ($\leq$50 s each)

1. Coefficient of x^3 in $(2+3x)^5$.

Ans: $\binom{5}{3}2^2 3^3 = 10 \cdot 4 \cdot 27 = 1\,080$

2. Coefficient of x^5 in $(1+2x)^4(1+x)^3$.

Ans: 168

3. Term independent of x in $\left(2x + \frac{1}{x^2}\right)^6$.

Ans: $\binom{6}{2}2^4 = 240$

4. Coefficient of x^2 in $(1+x)^n$ is 66. Find n.

Ans: $n = 12$

5. Coefficient of x^6 in $\left(x^2 + \frac{1}{x}\right)^6$.

Ans: $\binom{6}{2} = 15$

6. Coefficient of x^2 in $(1+2x)^4(1-x)^2$.

Ans: 9

7. Evaluate 11^3 and 9^3 by binomial.

Ans: $11^3 = 1331$; $9^3 = 729$

8. Coefficient of x^3 in $(1+kx)^6$ is 160. Find k.

Ans: $k = 2$

9. $\binom{n}{2} = \binom{n}{3}$. Find n.

Ans: $n = 5$

10. Coefficient of a^2b^2c in $(a+b+c)^5$. *(cf. TMUA 2020 P1 Q13)*

Ans: $\frac{5!}{2!2!1!} = 30$

Section C Substitution & Structure ($\leq$90 s each)

1. $\binom{10}{0} + \binom{10}{2} + \cdots + \binom{10}{10}$.

Ans: $2^9 = 512$

2. Greatest coefficient in $(1+x)^8$.

Ans: $\binom{8}{4} = 70$

3. Estimate 1.01^{10} from three terms.

Ans: $1 + 10(0.01) + 45(0.01)^2 = 1.1045$

4. In $(1 + 3x)^n$, coefficient of x^5 is three times that of x^4. Find n.

Ans: $n = 9$

5. Coefficient of x^4 in $(1 + x)^5(1 - 2x)^3$. *(cf. MAT 2023 Q1F)*

Ans: 25

6. $\displaystyle\sum_{k=0}^{5} k\binom{5}{k}$.

Ans: $5 \cdot 2^4 = 80$

7. Evaluate $\displaystyle\sum_{k=0}^{6} \binom{6}{k} 2^k$ by substitution.

Ans: $3^6 = 729$

8. Prove Pascal's rule by committee; build row 7 from row 6.

Ans: Row 7: $1, 7, 21, 35, 35, 21, 7, 1$

Section D Challenge (SMC / BMO1 difficulty)

1. Prove $\binom{2n}{n}$ is even for all $n \geq 1$.

Ans: Proof below.

2. Show $\displaystyle\sum_{k=0}^{10}\binom{10}{k}^2=\binom{20}{10}$.

Ans: Both count $\binom{20}{10}=184\,756$.

3. Prove exactly four of the $\binom{12}{k}$ are odd. *(cf. TMUA 2023 P1 Q6)*

Ans: The odd ones are exactly $k=0,4,8,12$.

4. Show $\binom{2}{2} + \binom{3}{2} + \cdots + \binom{9}{2} = \binom{10}{3}$.

Ans: $\binom{10}{3} = 120$

5. p prime, $0 < k < p$: prove $p \mid \binom{p}{k}$.

Ans: Proof below.